

PRANAV RAMESH BIDARE

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EDUCATION

Master of Science in Robotics and Autonomous Systems (Thesis Based MS)
Arizona State University, Tempe, AZ

December 2025
4.0/4.0 GPA

SKILLS

- **Programming:** Python, Matlab, C++
- **Frameworks/Tools:** ROS, ROS2, Autoware, CARLA, TensorFlow, PyTorch, OpenCV, Keras, Git, Jira
- **Concepts:** End-to-End Learning, 3D Perception Pipelines, Sensor Fusion, Sensor Calibration, Kalman Filters, Robotics, Machine Learning, Systems Engineering, Artificial Intelligence, Deep Learning, 3D Object Detection

EXPERIENCE

Researcher, BELIV Lab, Arizona State University

January 2024 - Present

- Optimized and deployed Autoware SceneSeg on-vehicle using ONNX and TensorRT, validated robustness across public ADAS hazard testing cases, including animal-related scenarios, in collaboration with the Autoware Foundation.
- Identified gaps in existing cooperative perception approaches, developed a real-time LiDAR-camera sensor fusion pipeline within a V2I late-fusion framework, extending detection range by 252% and reducing RMSE by 43%
- Established a distributed computing architecture by offloading GPU-intensive perception tasks to a Jetson Orin, reducing system latency and improving frame-rate to 20 fps for real-time object detection

Project Trainee, Bosch Global Software Technologies

January 2023 - May 2023

- Worked closely with senior engineers to design and prototype a proof of concept for Delta Firmware Over-the-Air (FOTA) updates using Diff-and-Patch algorithms which led to a reduction in vehicle data consumption
- Collaborated with a cross-functional Agile team to develop Connected Vehicle Technology, addressing diverse customer requirements while leveraging JIRA for sprint planning and Git for version control

Research Intern, CORI Lab, PES University

January 2022 - January 2023

- Teamed up to design and implement multiple Generative Adversarial Network (GAN) models, including a novel architecture for low-light vehicle detection that improved accuracy by 28%, a real-time fire detection system for space capsules achieving 95% accuracy, and a spoken digit recognition model reaching 99% accuracy

PROJECTS

Vehicle-to-Infrastructure (V2I) Based 3D Object Detection and Sensor Fusion

- Built an end-to-end sensor fusion perception stack in Python and ROS2 for 3D object detection, merging onboard LiDAR with infrastructure camera data
- Designed a dual-update Kalman Filter to fuse asynchronous LiDAR and camera data, overcoming network delays and noise to deliver smoother, more reliable multi-sensor state estimation
- Created and executed 4 different urban traffic scenarios using CARLA and Scenario Runner and conducted real-time validation of the proposed pipeline using the Ego Vehicle and an RSU Camera setup

Multi Camera-LiDAR Calibration for Multi-View Perception

- Led a system upgrade from a 135° forward-facing setup to full 360° surround perception using a 6-camera, 2-LiDAR sensor setup
- Integrated cameras and LiDAR into the Autoware ROS2 pipeline and performed intrinsic and extrinsic calibration using AprilTag-based PnP and SQPnP optimization, enabling accurate spatial alignment for multi-sensor fusion

Traffic Light and Sign Detection for Autonomous Driving in Challenging Environments

- Engineered a YOLOv11n-based detection pipeline in TensorFlow, trained on LISA and CARLA-augmented data with motion blur, fog, rain, and occlusions to improve model robustness
- Achieved 89.3% mAP@50 on the LISA benchmark and retained 78.1% mAP@50 across augmented tests, demonstrating high resilience to environmental perturbations

RESEARCH PAPERS

- “Optimizing Perception Capabilities of Autonomous Vehicles Through V2I Late Fusion using Kalman Filter”, IFAC MECC 2025
- “Upgrading an Automated Vehicle Research Platform for Enhanced Perception and Distributed Computing”, IFAC IAV 2025
- “BaTCAVE: Trustworthy Explanations for Robot Behaviors”, IEEE/RSJ IROS, 2025
- “Design of a Real-Time GAN-based Speech Recognizer for Consumer Electronics”, IEEE ICICT 2023
- “Design and Evaluation of a Vehicle Detection System in Low Light Conditions”, IEEE SPIN 2023
- “Design and Evaluation of an E-Eye for Fire Detection in Human Space Capsule”, IEEE ICARES 2022